

been early-deprecated, as we can no longer access it despite the supposed June 13, 2024 deprecation date on <https://platform.openai.com/docs/deprecations>.

Access dates by figure in main text:

- gpt-3.5-turbo-0301, Figure 4: April 7, 2023
- gpt-3.5-turbo-0301, Figure 5: May 18, 2023
- gpt-3.5-turbo-0301, Figure 6 left two columns: May 18, 2023
- gpt-3.5-turbo-0301, Figure 6 right two columns: April 7, 2023
- gpt-3.5-turbo-0301, Figure 7-8: January 25, 2024
- gpt-3.5-turbo-0301, Figure 10-11 May 24, 2024
- gpt-4-0314, Figure 12: May 2, 2023
- gpt-3.5-turbo-0613, Figure 12-13: January 25, 2024
- gpt-3.5-turbo-1106, Figure 12-13: January 29, 2024
- gpt-4-0613, Figure 12-13: January 25, 2024
- gpt-4-1106-preview, Figure 12-13: January 29, 2024

E.3. Example prompts

In this section, we provide example prompts we used for various experiments. For simplicity, we use the same query for each prompt shown below, and we only use 2 shots (most experiments in the main text are done with 8 shots). In practice, we sampled shots randomly (controlling for digit length to match the query length) for each query, as explained in Section 2.1. For the experiments described in Section 4 and Appendix E.1, the shots were also controlled to have the same answer length as the query. The examples we present below, though, are for the runs in the rest of the paper (where only input digit lengths are controlled). For maximum clarity, we display prompts as the list of dictionaries that gets sent to OpenAI’s API and roughly in the order used for figures in the paper. Following the advice at <https://platform.openai.com/docs/guides/prompt-engineering/tactic-provide-examples>, we make use of the multi-turn chat dialog to prompt the model, as opposed to one big user message with all the examples.

L2R tokenization, input-digit-controlled for two 7-digit numbers:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3790206+6739555='},
{'role': 'assistant', 'content': '10529761'},
{'role': 'user', 'content': '6777159+7096168='},
{'role': 'assistant', 'content': '13873327'},
{'role': 'user', 'content': '8302080+3529456='}]
```

R2L tokenization, input-digit-controlled for two 7-digit numbers:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3,790,206+6,739,555='},
{'role': 'assistant', 'content': '10,529,761'},
{'role': 'user', 'content': '6,777,159+7,096,168='},
{'role': 'assistant', 'content': '13,873,327'},
{'role': 'user', 'content': '8,302,080+3,529,456='}]
```

R2L tokenization, delimiter-control condition using '#':

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3#790#206+6#739#555='},
{'role': 'assistant', 'content': '10#529#761'},
{'role': 'user', 'content': '6#777#159+7#096#168='},
{'role': 'assistant', 'content': '13#873#327'},
{'role': 'user', 'content': '8#302#080+3#529#456='}]
```

L2R tokenization, thinking token control by using separators in L2R tokenization:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '379,020,6+673,955,5='},
{'role': 'assistant', 'content': '105,297,61'},
{'role': 'user', 'content': '677,715,9+709,616,8='},
{'role': 'assistant', 'content': '138,733,27'},
{'role': 'user', 'content': '830,208,0+352,945,6='}]
```

L2R tokenization, thinking token control by using 2 extra spaces:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3790206 + 6739555 = '},
{'role': 'assistant', 'content': '10529761'},
{'role': 'user', 'content': '6777159 + 7096168 = '},
{'role': 'assistant', 'content': '13873327'},
{'role': 'user', 'content': '8302080 + 3529456 = '}]
```

L2R tokenization, thinking token control by using 2 extra spaces:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3790206 + 6739555 = '},
{'role': 'assistant', 'content': '10529761'},
{'role': 'user', 'content': '6777159 + 7096168 = '},
{'role': 'assistant', 'content': '13873327'},
{'role': 'user', 'content': '8302080 + 3529456 = '}]
```

Repeat L2R → R2L:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3790206+6739555='},
{'role': 'assistant', 'content': '3,790,206+6,739,555=10,529,761'},
{'role': 'user', 'content': '6777159+7096168='},
{'role': 'assistant', 'content': '6,777,159+7,096,168=13,873,327'},
{'role': 'user', 'content': '8302080+3529456='}]
```

Repeat control L2R → L2R:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},
{'role': 'user', 'content': '3790206+6739555='},
{'role': 'assistant', 'content': '3790206+6739555=10529761'},
{'role': 'user', 'content': '6777159+7096168='},
{'role': 'assistant', 'content': '6777159+7096168=13873327'},
{'role': 'user', 'content': '8302080+3529456='}]
```

Output control L2R $\rightarrow$ R2L:

```
[{'role': 'system', 'content': 'You are a helpful assistant.'},  
{ 'role': 'user', 'content': '3790206+6739555='},  
{ 'role': 'assistant', 'content': '10,529,761'},  
{ 'role': 'user', 'content': '6777159+7096168='},  
{ 'role': 'assistant', 'content': '13,873,327'},  
{ 'role': 'user', 'content': '8302080+3529456='}]
```